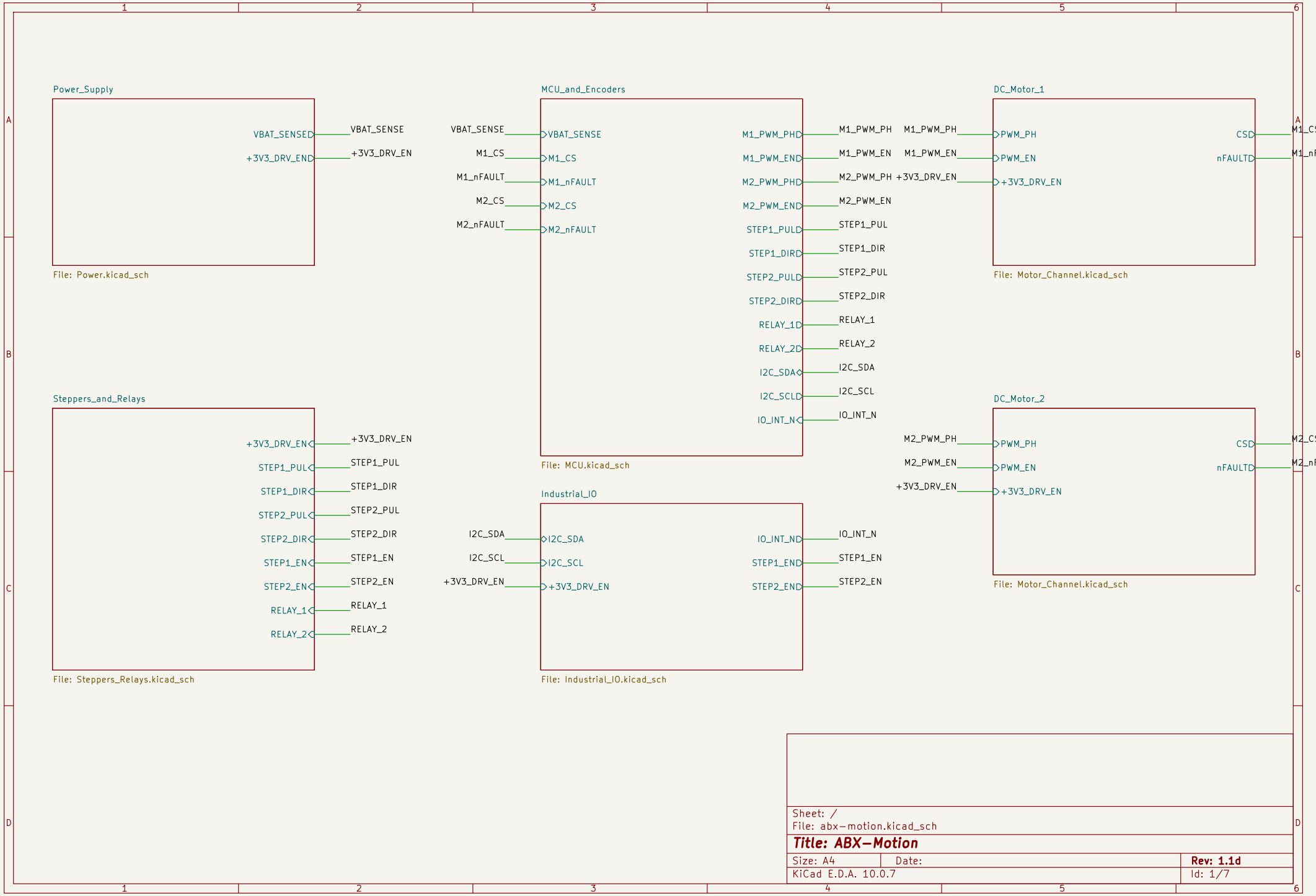
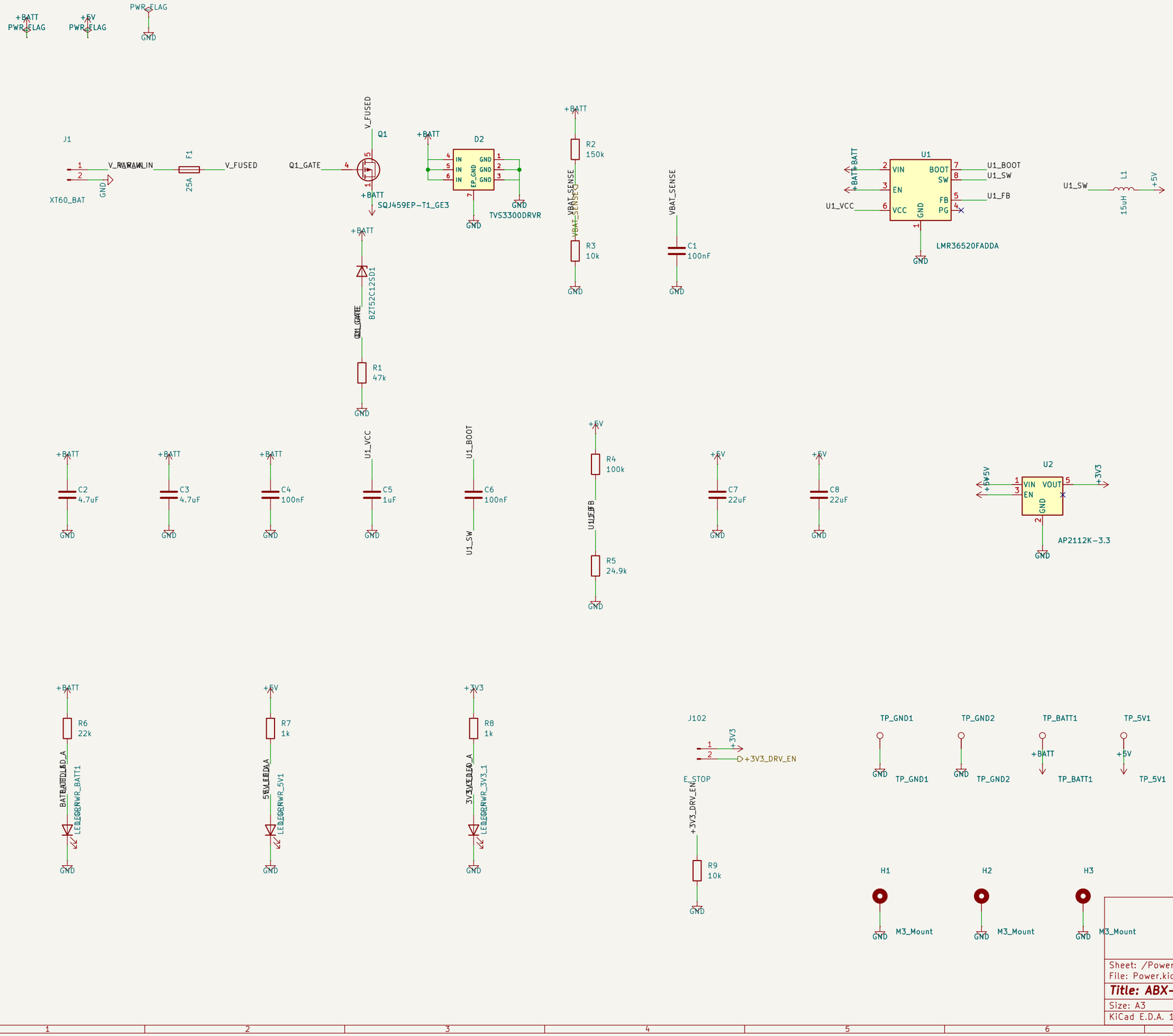

Power_Supply
VBAT_SENSE
+3V3_DRV_EN
VBAT_SENSE
+3V3_DRV_EN
File: Power.kicad_sch
MCU_and_Encoders
VBAT_SENSE
M1_CS
M1_nFAULT
M2_CS
M2_nFAULT
M1_PWM_PH
M1_PWM_EN
M2_PWM_PH
M2_PWM_EN
STEP1_PUL
STEP1_DIR
STEP2_PUL
STEP2_DIR
RELAY_1
RELAY_2
I2C_SDA
I2C_SCL
IO_INT_N
File: MCU.kicad_sch
DC_Motor_1
M1_PWM_PH
M1_PWM_EN
+3V3_DRV_EN
CSD
nFAULTD
M1_CS
M1_nFAULT
File: Motor_Channel.kicad_sch
Steppers_and_Relays
+3V3_DRV_EN
STEP1_PUL
STEP1_DIR
STEP2_PUL
STEP2_DIR
STEP1_EN
STEP2_EN
RELAY_1
RELAY_2
File: Steppers_Relays.kicad_sch
Industrial_IO
I2C_SDA
I2C_SCL
+3V3_DRV_EN
IO_INT_N
STEP1_EN
STEP2_EN
File: Industrial_IO.kicad_sch
DC_Motor_2
M2_PWM_PH
M2_PWM_EN
+3V3_DRV_EN
CSD
nFAULTD
M2_CS
M2_nFAULT
File: Motor_Channel.kicad_sch
Sheet: /
File: abx-motion.kicad_sch
Title: ABX-Motion
Size: A4
Date:
Rev: 1.1d
KiCad E.D.A. 10.0.7
Id: 1/7

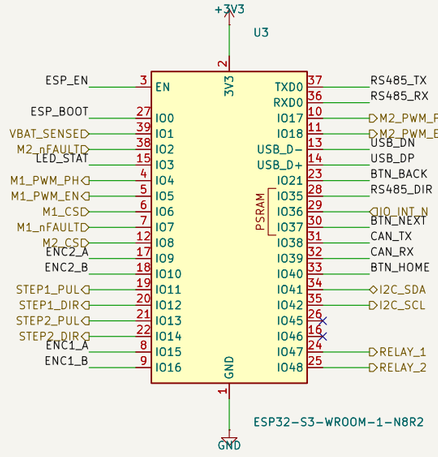
SHEET 1: BATTERY INPUT, REVERSE POLARITY PROTECTION, 5V BUCK & 3.3V LDO

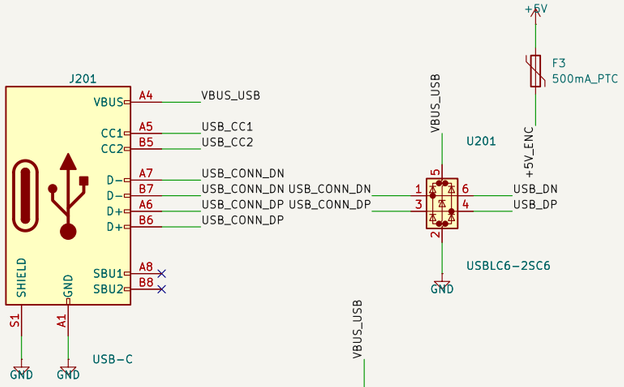
D2: 33V standoff; 40V max at specified 8/20us pulse. No continuous braking-energy rating.

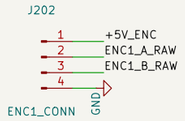


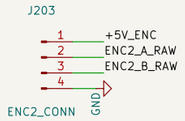
Sheet: /Power_Supply/		
File: Power.kicad_sch		
Title: ABX-Motion		
Size: A3	Date:	**Rev: 1.1d**
KiCad E.D.A. 10.0.7		Id: 2/7

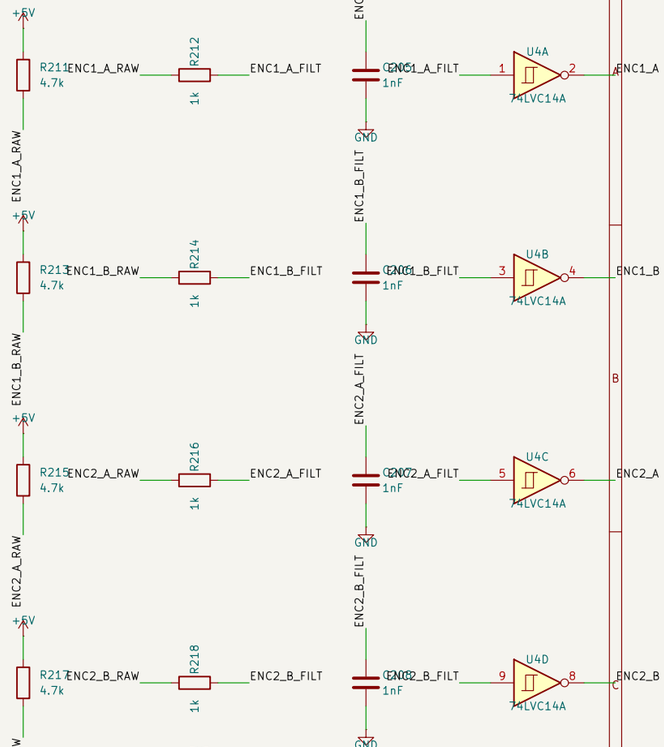
SHEET 2: ESP32-S3-WROOM-1, USB-C, 74LVC14A ENCODERS, CAN, RS-485, OLED & UI BUTTONS

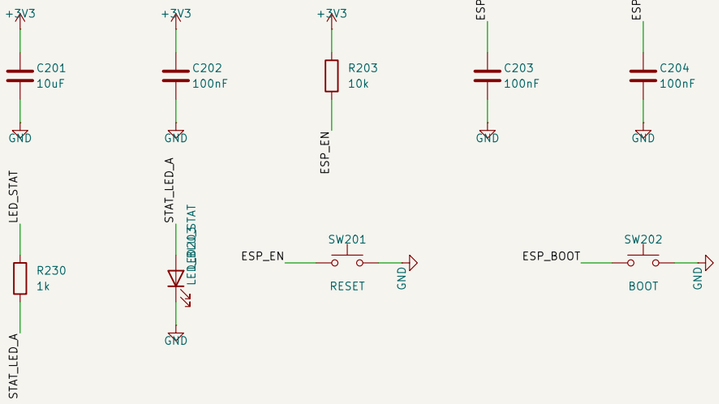


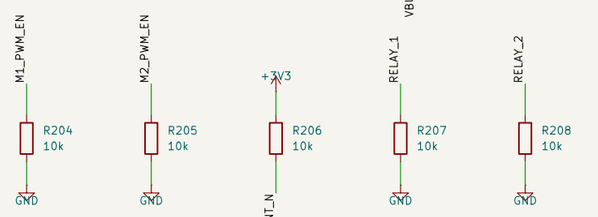


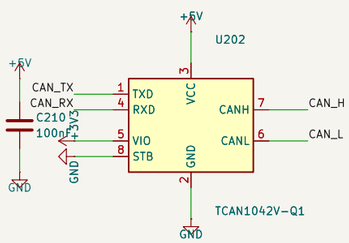


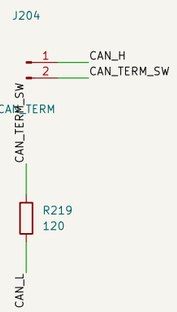


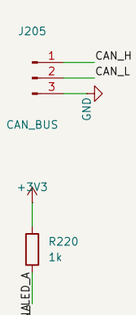


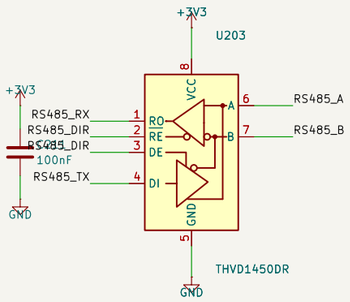


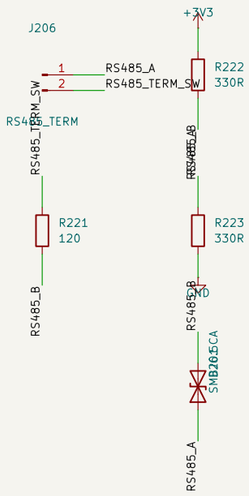




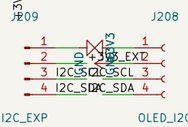


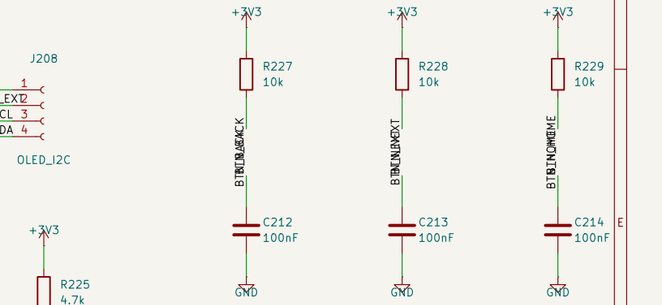


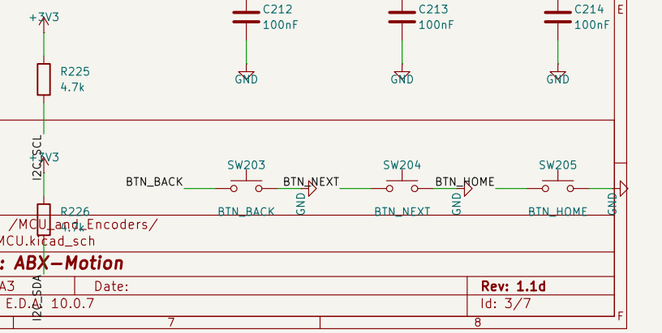










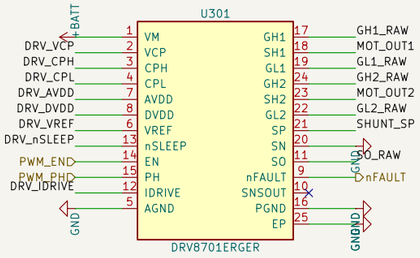


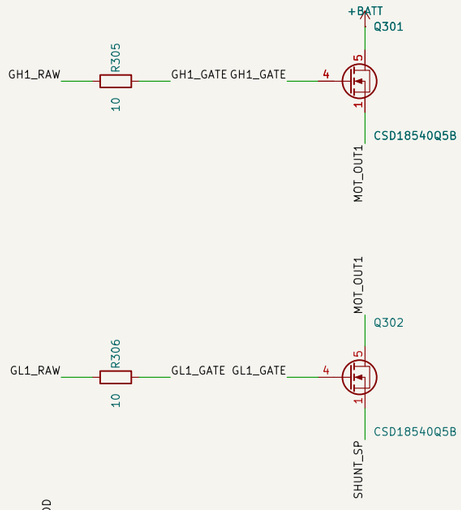
Sheet: /MCU and Encoders/
File: MCU.kicad_sch

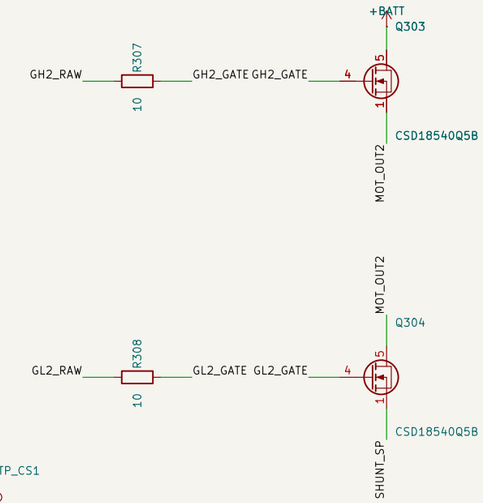
Title: ABX-Motion

Size: A3	Date:	**Rev: 1.1d**
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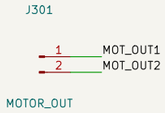
SHEET 3: 100W BRUSHED DC SERVO CHANNEL (DRV8701ERGER + 4x CSD18540Q5B 60V H-BRIDGE)

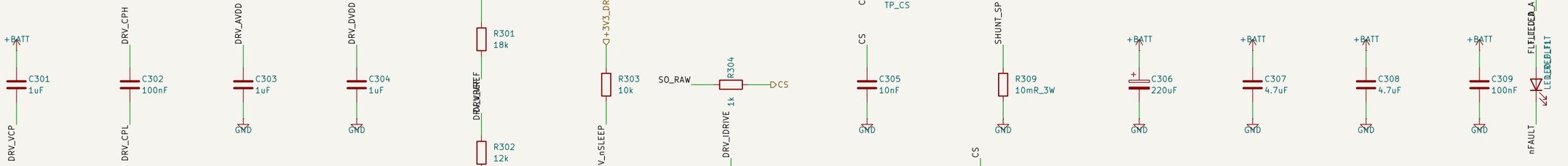








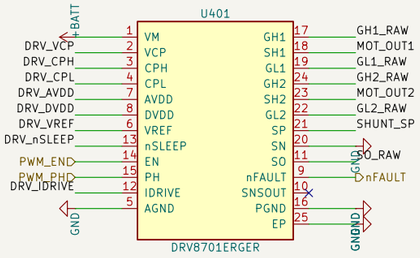


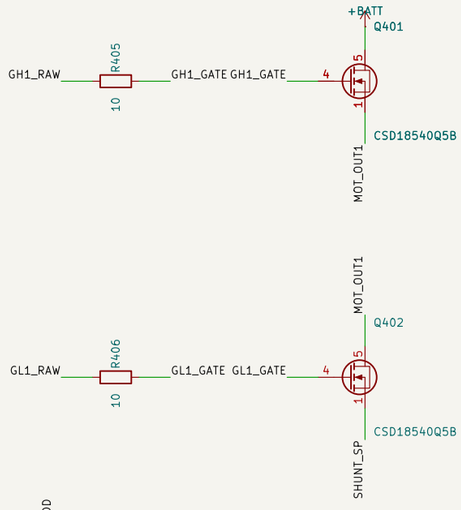


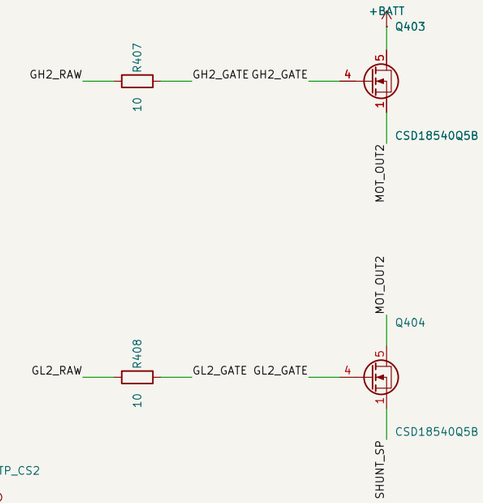
ADC protection: R304/R404 + R313/R413 divide SO by 2; rescale firmware.

Sheet: /DC_Motor_1/		
File: Motor_Channel.kicad_sch		
Title: ABX-Motion		
Size: A3	Date:	**Rev: 1.1d**
KiCad E.D.A. 10.0.7		Id: 4/7

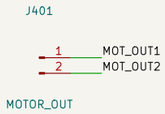
SHEET 3: 100W BRUSHED DC SERVO CHANNEL (DRV8701ERGER + 4x CSD18540Q5B 60V H-BRIDGE)











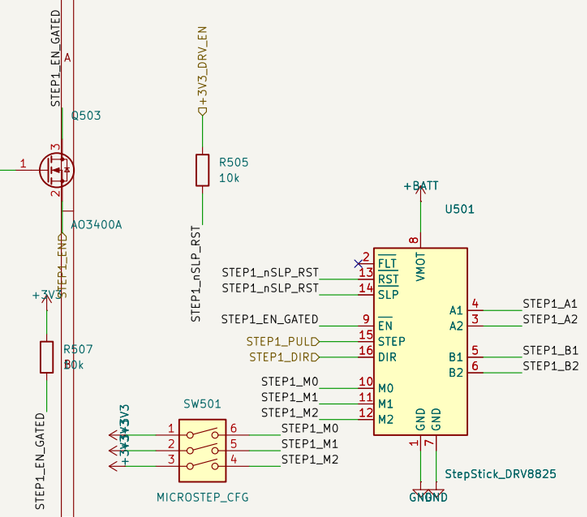


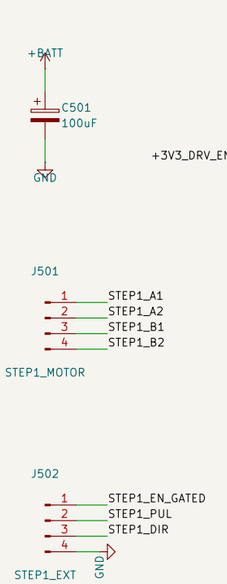


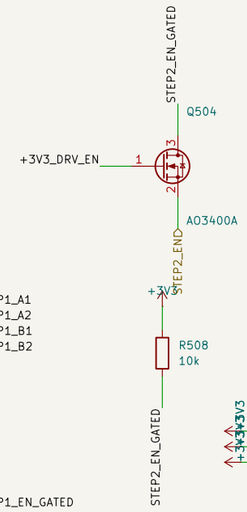
ADC protection: R304/R404 + R313/R413 divide SO by 2; rescale firmware.

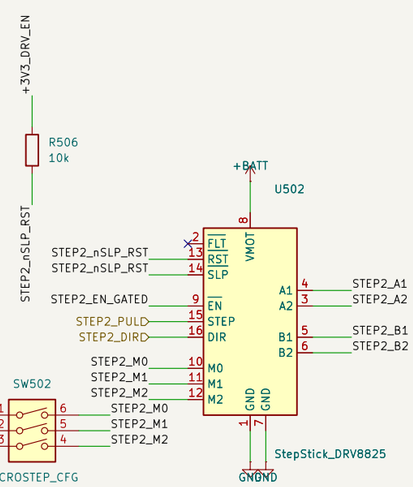
Sheet: /DC_Motor_2/		
File: Motor_Channel.kicad_sch		
Title: ABX-Motion		
Size: A3	Date:	**Rev: 1.1d**
KiCad E.D.A. 10.0.7		Id: 5/7

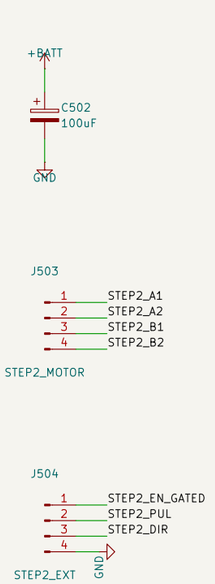
SHEET 4: DUAL STEPSTICK STEPPER CHANNELS & DUAL 5V OMRON G5LE RELAY OUTPUTS

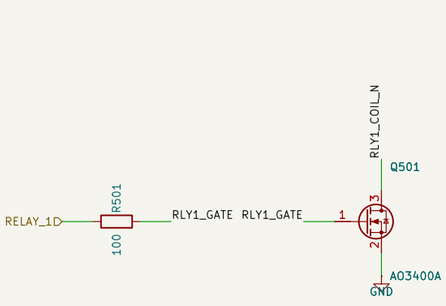




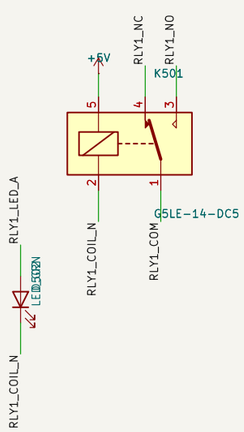


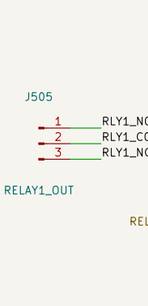


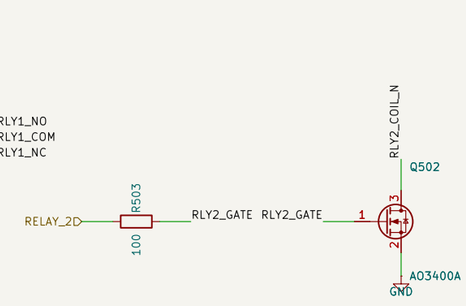




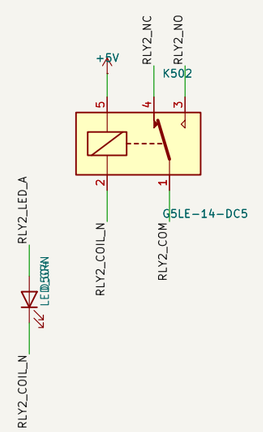


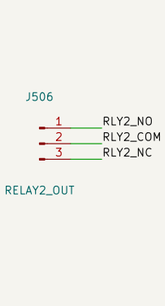












Sheet: /Steppers_and_Relays/
File: Steppers_Relays.kicad_sch

Title: ABX—Motion

Size: A3	Date:	**Rev: 1.1d**
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SHEET 5: 6x 3-PIN JST-PH INDUSTRIAL I/O (8x LIMIT/GPIO + 2x 5V/500mA DC OUTPUTS) & PCA9555PW EXPANDER

J601
B3B-PH-K-S
M1_MIN_RAW
M1_MAX_RAW
R601 4.7k
R602 4.7k
R609 1k
R610 1k
C602 10nF
C603 10nF

J602
B3B-PH-K-S
M2_MIN_RAW
M2_MAX_RAW
R603 4.7k
R604 4.7k
R611 1k
R612 1k
C604 10nF
C605 10nF

J603
B3B-PH-K-S
S1_MIN_RAW
S1_MAX_RAW
R605 4.7k
R606 4.7k
R613 1k
R614 1k
C606 10nF
C607 10nF

J604
B3B-PH-K-S
S2_MIN_RAW
S2_MAX_RAW
R607 4.7k
R608 4.7k
R615 1k
R616 1k
C608 10nF
C609 10nF

U601
PCA9555PW,118
I2C_SCL
I2C_SDA
IO_INT_N
C601 100nF
M1_MIN_IN
M1_MAX_IN
M2_MIN_IN
M2_MAX_IN
S1_MIN_IN
S1_MAX_IN
S2_MIN_IN
S2_MAX_IN
OUT1_CMD
OUT2_CMD
STEP1_EN
STEP2_EN
+3V3_DRV_EN

+5V
1206L050/16YR
+5V_OUT

OUT1_CMD
R617 1k
R619 100
OUT1_GATE
Q601
AO3400A
OUT1_SW
B5819W
J605
B3B-PH-K-S

OUT2_CMD
R618 1k
R620 100
OUT2_GATE
Q602
AO3400A
OUT2_SW
B5819W
J606
B3B-PH-K-S

R621 10k
R622 10k
STEP1_EN
STEP2_EN

Sheet: /Industrial_IO/
File: Industrial_IO.kicad_sch
Title: ABX-Motion
Size: A3 | Date:
KiCad E.D.A. 10.0.7
Rev: 1.1d
Id: 7/7